Exam
DIT008: Discrete Mathematics

University of Gothenburg
Lecturer: Jan Gerken

7 January 2026, 8:30 – 12:30

No calculators or other aids are allowed.

You must show your calculations or describe your argument. Unless otherwise stated, if you provide just the final answer, you will not be awarded any points.

The grade boundaries are

- pass (3): at least 20 points (50%)
- pass with credit (4): at least 28 points (70%)
- distinction (5): at least 36 points (90%)

In case of questions, call Jan Gerken at 031-772-14-37.

Good luck!

Problem:	1	2	3	4	5	6	7	8	9	Total
Points:	5	2	4	4	5	5	8	3	4	40

Problem 1 (5 points)
Determine whether the statement forms

$$p \wedge (q \vee r) \quad \text{and} \quad (p \wedge q) \vee (p \wedge r)$$

are logically equivalent. Construct a truth table and include a sentence justifying your answer. Your sentence should show that you understand the meaning of logical equivalence.

Solution:

p	q	r	$q \vee r$	$p \wedge q$	$p \wedge r$	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	F	T	T
T	F	T	T	F	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

Since the last two columns have the same truth value for all values of the statement variables, the two statements $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$ are equivalent.

Problem 2 (2 points)
Given subsets A and B of a universal set U, provide the definitions for the following in terms of the elements of A and B. No justification needed, 0.5 points each.

(a) $A \subseteq B$

(b) $A \cup B$

(c) $A \cap B$

(d) $A - B$

Solution:

(a) $A \subseteq B \quad \Leftrightarrow \quad x \in A \Rightarrow x \in B.$

(b) $A \cup B = \{x \in U \mid x \in A \text{ or } x \in B\}$

(c) $A \cap B = \{x \in U \mid x \in A \text{ and } x \in B\}$

(d) $A - B = \{x \in U \mid x \in A \text{ and } x \notin B\}$

Problem 3 (4 points)
Prove the following statement using an element argument and reasoning directly from the definitions.

$$\text{For all sets } A, B \text{ and } C, (A \cup B) \cap C \subseteq A \cup (B \cap C).$$

Solution:
Proof: Let $x \in (A \cup B) \cap C$ be arbitrary. By definition of intersection, we have $x \in A \cup B$ and $x \in C$.

Since $x \in A \cup B$, by definition of union, we have $x \in A$ or $x \in B$.

Case 1: Suppose $x \in A$. Then $x \in A \cup (B \cap C)$ by definition of union.

Case 2: Suppose $x \in B$. Since we also have $x \in C$, by definition of intersection we get $x \in B \cap C$. Therefore, $x \in A \cup (B \cap C)$ by definition of union.

In both cases, we have shown that $x \in A \cup (B \cap C)$.

Since x was arbitrary, we conclude that $(A \cup B) \cap C \subseteq A \cup (B \cap C)$.

Problem 4 (4 points)
Prove by induction that

$$\sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2 \qquad \forall\, n \geq 1\,.$$

Solution: Let the property $P(n)$ be the equation

$$\sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2 . \tag{$P(n)$}$$

We will show that $P(n)$ is true for every integer $n \geq 1$.

Base case ($n = 1$): $P(1)$ is true because the left-hand side is

$$\sum_{i=1}^{1} i^3 = 1^3 = 1$$

and the right-hand side is

$$\left(\frac{1(1+1)}{2}\right)^2 = \left(\frac{2}{2}\right)^2 = 1^2 = 1$$

also.

Inductive step: Let k be any integer with $k \geq 1$, and suppose that $P(k)$ is true, i.e. that

$$\sum_{i=1}^{k} i^3 = \left(\frac{k(k+1)}{2}\right)^2 . \tag{$P(k)$}$$

We must show that

$$\sum_{i=1}^{k+1} i^3 = \left(\frac{(k+1)(k+2)}{2}\right)^2 . \tag{$P(k+1)$}$$

The left-hand side of $P(k+1)$ is

$$\begin{aligned}
\sum_{i=1}^{k+1} i^3 &= \sum_{i=1}^{k} i^3 + (k+1)^3 \\
&= \left(\frac{k(k+1)}{2}\right)^2 + (k+1)^3 \quad \text{by inductive hypothesis} \\
&= \frac{k^2(k+1)^2}{4} + (k+1)^3 \\
&= (k+1)^2 \left(\frac{k^2}{4} + (k+1)\right) \\
&= (k+1)^2 \left(\frac{k^2+4k+4}{4}\right) \\
&= (k+1)^2 \left(\frac{(k+2)^2}{4}\right) \\
&= \left(\frac{(k+1)(k+2)}{2}\right)^2 ,
\end{aligned}$$

which is the right-hand side of $P(k+1)$ as was to be shown.

Problem 5 (5 points)

A game involves placing tokens on a row of n squares. Each square can either be empty, contain one red token, or contain one blue token. However, two adjacent squares cannot both contain blue tokens. Let g_n be the number of different ways to fill a row of n squares following this rule.

(a) (1 point) Find g_1 and g_2. Justify your answer.

(b) (3 points) Find a recurrence relation for g_n where $n \geq 3$. Justify your answer carefully.

(c) (1 point) Use your recurrence relation to compute g_3 and g_4.

Solution:

(a) $g_1 = 3$ (the square can be empty, contain a red token, or contain a blue token)

$g_2 = 8$ (the valid configurations are: EE, ER, EB, RE, RR, RB, BE, BR; note that BB is not allowed, where E = empty, R = red, B = blue)

(b) For $n \geq 3$, consider a valid configuration of n squares. We examine what is in the n-th (rightmost) square:

- If the n-th square is empty, the first $n-1$ squares can be filled in g_{n-1} ways.
- If the n-th square contains a red token, the first $n-1$ squares can be filled in g_{n-1} ways.
- If the n-th square contains a blue token, the $(n-1)$-th square cannot contain a blue token, so it must be either empty or contain a red token (2 choices). The first $n-2$ squares can be filled in g_{n-2} ways, giving $2 \cdot g_{n-2}$ configurations.

Therefore, for all integers $n \geq 3$,

$$g_n = g_{n-1} + g_{n-1} + 2g_{n-2} = 2g_{n-1} + 2g_{n-2}\,.$$

(c) Using the recurrence relation:

$$\begin{aligned} g_3 &= 2g_2 + 2g_1 = 2(8) + 2(3) = 16 + 6 = 22 \\ g_4 &= 2g_3 + 2g_2 = 2(22) + 2(8) = 44 + 16 = 60 \end{aligned}$$

Problem 6 (5 points)

A committee of 8 people consists of 5 teachers and 3 administrators. A subcommittee of 4 people is to be formed.

(a) (1 point) How many different subcommittees of 4 people can be formed?

(b) (1 point) How many different subcommittees contain exactly 3 teachers and 1 administrator?

(c) (1 point) How many different subcommittees contain at least 2 administrators?

(d) (2 points) If two specific teachers refuse to serve on the same subcommittee (but each is willing to serve if the other does not), how many different subcommittees can be formed?

Solution:

(a) There are

$$\binom{8}{4} = \frac{8!}{4! \cdot 4!} = \frac{8 \cdot 7 \cdot 6 \cdot 5}{4 \cdot 3 \cdot 2} = 2 \cdot 7 \cdot 5 = 70$$

different subcommittees of 4 that can be formed.

(b) Choose 3 teachers from 5 and 1 administrator from 3:

$$\binom{5}{3} \cdot \binom{3}{1} = 10 \cdot 3 = 30\,.$$

(c) "At least 2 administrators" means exactly 2 or exactly 3 administrators.

Exactly 2 administrators: $\binom{3}{2} \cdot \binom{5}{2} = 3 \cdot 10 = 30$

Exactly 3 administrators: $\binom{3}{3} \cdot \binom{5}{1} = 1 \cdot 5 = 5$

Total: $30 + 5 = 35$

(d) Call the two teachers T_1 and T_2. We count three cases:

Case 1: T_1 serves but not T_2. Choose 3 more from the remaining 6 people: $\binom{6}{3} = 20$

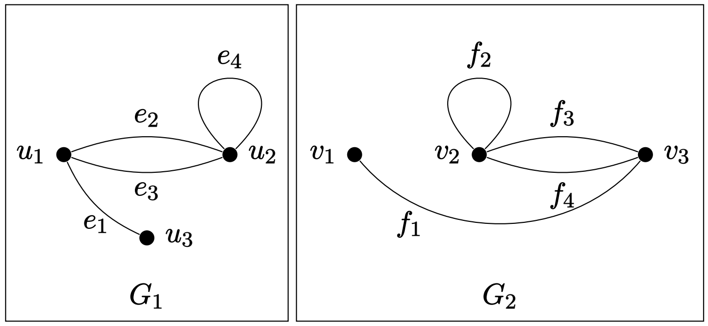
Case 2: T_2 serves but not T_1. Choose 3 more from the remaining 6 people: $\binom{6}{3} = 20$

Case 3: Neither serves. Choose 4 from the remaining 6 people: $\binom{6}{4} = 15$

Total: $20 + 20 + 15 = 55$

Problem 7 (8 points)

Consider the following graphs G_1 and G_2



(a) (2 points) Compute the adjacency matrix A_1 for G_1 with vertex ordering (u_1, u_2, u_3) and the adjacency matrix A_2 for G_2 with vertex ordering (v_1, v_2, v_3).

(b) (2 points) G_1 and G_2 are isomorphic. Find the vertex function $g : V(G_1) \to V(G_2)$ and edge function $h : E(G_1) \to E(G_2)$ that define the isomorphism.

(c) (2 points) Let P be the 3×3 matrix defined by

$$P_{ij} = \begin{cases} 1 \text{ if } g(u_i) = v_j \\ 0 \text{ otherwise} \end{cases} .$$

Write down P and $P^\top$, where $P^\top$ is the *transpose* of P with entries $(P^\top)_{ij} = P_{ji}$.

(d) (2 points) Compute $P^\top A_1 P$. If you did not find A_1 and P above, you may use

$$A_1 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 2 & 0 & 2 \end{pmatrix} \qquad P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} .$$

Solution:

(a) The adjacency matrices are

$$A_1 = \begin{pmatrix} 0 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \qquad A_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{pmatrix} .$$

(b) The functions can be specified by providing their values on the entire domain,

$$\begin{array}{llll} g(u_1) = v_3 & g(u_2) = v_2 & g(u_3) = v_1 & \\ h(e_1) = f_1 & h(e_2) = f_3 & h(e_3) = f_4 & h(e_4) = f_2 \,. \end{array}$$

Note that $h(e_2) = f_4$ and $h(e_3) = f_3$ is also correct.

(c)

$$P = P^\top = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

(d) If the matrices A_1 and P defined by the graphs are used, we obtain

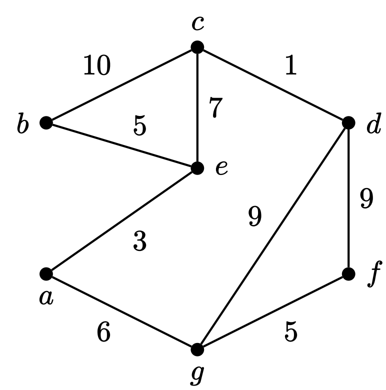
$$P^\top A_1 P = P^\top \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix} P = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{pmatrix} = A_2 .$$

For the alternative matrices A_1 and P, we obtain instead

$$P^\top A_1 P = P^\top \begin{pmatrix} 2 & 3 & 6 \\ 2 & 1 & 6 \\ 0 & 2 & 4 \end{pmatrix} = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 0 & 2 \\ 6 & 4 & 6 \end{pmatrix} P = \begin{pmatrix} 2 & 1 & 6 \\ 0 & 2 & 4 \\ 4 & 6 & 16 \end{pmatrix}.$$

Problem 8 (3 points)

For the following weighted graph, use Prim's algorithm starting with vertex a to find a minimum spanning tree. Indicate the order in which the edges are added to build the tree.



Solution: Order of adding the edges: $\{a,e\}, \{b,e\}, \{a,g\}, \{g,f\}, \{e,c\}, \{c,d\}$

Problem 9 (4 points)
Prove that $3x^3 + 5x + 2$ is $\Theta(x^3)$ without using the theorem on polynomial orders.

Solution: To show that $3x^3 + 5x + 2$ is $\Theta(x^3)$, we must show that it is both $O(x^3)$ and $\Omega(x^3)$.

Part 1: Show that $3x^3 + 5x + 2$ is $O(x^3)$

$\forall\ x > 1$,

$$0 \le 3x^3 + 5x + 2$$

because $3x^3$, $5x$, and 2 are all positive for $x > 1$. Furthermore,

$$3x^3 + 5x + 2 \le 3x^3 + 5x^3 + 2x^3 = 10x^3$$

because $x < x^3$ and $1 < x^3$ for $x > 1$. Therefore, we have

$$0 \le 3x^3 + 5x + 2 \le 10x^3 \quad \forall\ x > 1\,.$$

Hence, by definition of O-notation, $3x^3 + 5x + 2$ is $O(x^3)$.

Part 2: Show that $3x^3 + 5x + 2$ is $\Omega(x^3)$

$\forall\ x > 1$,

$$3x^3 + 5x + 2 \ge 3x^3$$

because $5x$ and 2 are both positive for $x > 1$. Therefore, we have

$$3x^3 + 5x + 2 \ge 3x^3 > 0 \quad \forall\ x > 1\,.$$

Hence, by definition of Ω-notation, $3x^3 + 5x + 2$ is $\Omega(x^3)$.

Conclusion: Since $3x^3 + 5x + 2$ is both $O(x^3)$ and $\Omega(x^3)$, it follows that $3x^3 + 5x + 2$ is $\Theta(x^3)$.